Scalable, adaptive, and explainable scientific machine learning with applications to surrogate models of partial differential equations

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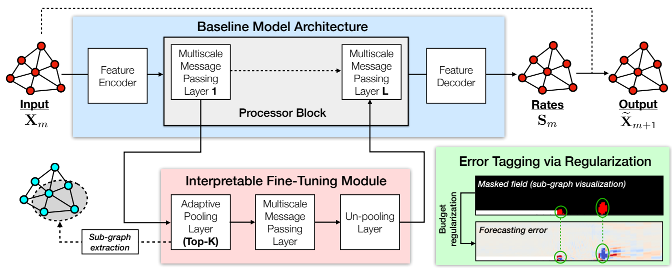
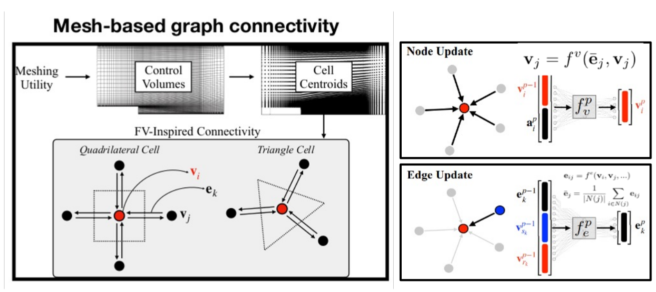
- **Motivation** – Interpretability of deep learning method is particularly important in engineering applications, where access to failure modes and model-form errors can prevent potentially high-cost catastrophic device or system operation failures [24].
- **Goal** – In this study, our goal is to introduce a scalable, adaptive and interpretable fine-tuning strategy for graph neural networks (GNNs), with application to unstructured mesh-based fluid dynamics datasets. This fine-tuned model isolates regions in the physical space, which are adaptively produced in the forward pass and serve as explainable links between the baseline model architecture, the optimization goal, and known problem-specific physics.

Methodology

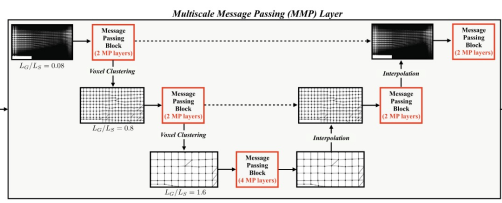
- Overall flowchart



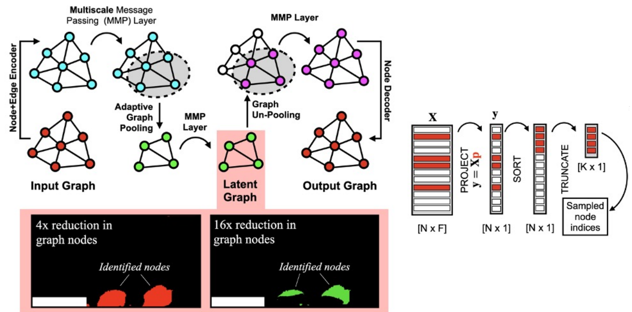
- Graph neural network (GNN) for unstructured data

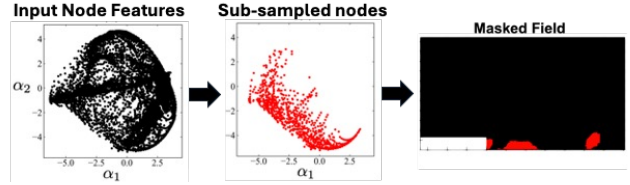


- Scalability via multiscale message passing



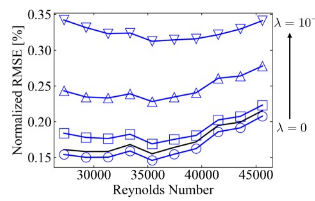
- Adaptability and Interpretability via adaptive subsampling

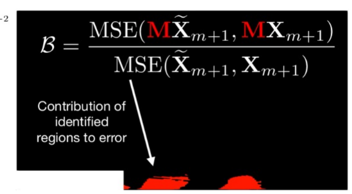


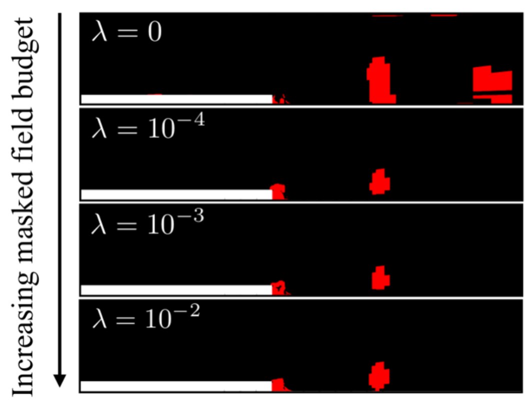


- Error-identifying GNNs by regularized fine-tuning objective:

$$\mathcal{L} = MSE(\tilde{X}_{m+1}, X_{m+1}) + \boxed{\lambda \cdot \frac{1}{\mathcal{B}}}$$



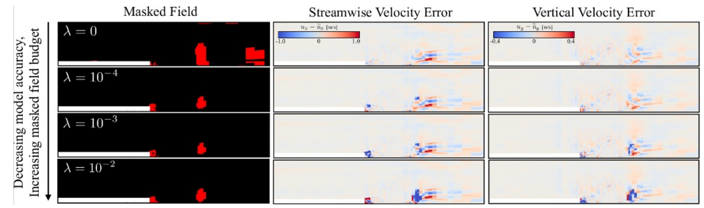


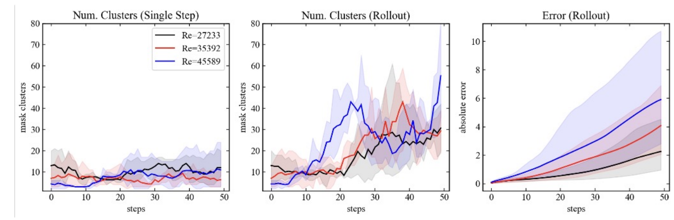


Masked fields move to accommodate regions of high errors

Results

- Latent space that tracks errors

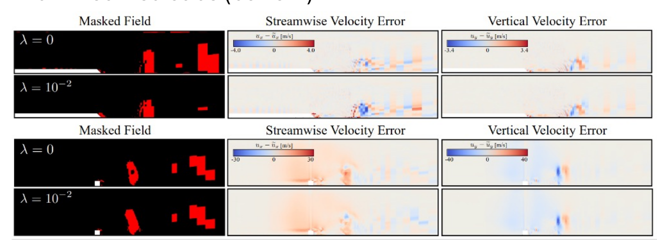




An HDBScan based clustering of subgraph coordinates of the latent-space features. the error grows and the number of clusters is seen to rise (along with the error). We can extrapolation in the absence of test data!

Generalization

- Separation off a smooth ramp (top) and flow-over a wall-mounted cube (bottom)



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